Week Summary

Numerical Solutions to PDEs

Numerical solutions to partial differential equations

- This week we looked at three differential equations:

1. The exponential decay equation: an ordinary differential equation with **time derivative**

2. The advection equation: a partial differential equation with a **time derivative** and a **first order spatial derivative**

3. The diffusion equation: a partial differential equation with a **time derivative** and a **second order spatial derivative**

Numerical solutions to partial differential equations

Exponential decay: $$\frac{dN}{dt} = -\lambda N$$

Advection: $$\frac{\partial u}{\partial t} = -c\frac{\partial u}{\partial x}$$

Diffusion: $$\frac{\partial u}{\partial t} = K\frac{\partial^2 u}{\partial x^2}$$

Numerical solutions to partial differential equations

Exponential decay: $$\boxed{\frac{dN}{dt}} = -\lambda N$$

Advection: $$\boxed{\frac{\partial u}{\partial t}} = \boxed{-c\frac{\partial u}{\partial x}}$$

Diffusion: $$\boxed{\frac{\partial u}{\partial t}} = \boxed{K\frac{\partial^2 u}{\partial x^2}}$$

Calculate with a finite difference stencil

Solve by time stepping

Accuracy and stability

- The finite difference representation has a **truncation error**
- The truncation error depends on the type of stencil used and the spacing of the grid points (**spatial resolution**)
- Better spatial resolution (more grid points) reduces the truncation error but is more computationally expensive
- Our scheme for solving the advection equation is stable provided the time step is not too large (**CFL condition**)
- The time step constraint also affects the computational cost of calculating the solution